Sheet #1 Dr. Negm Shawky Distributed Systems 1 October 2026

Answer all the questions

1. What is the simple **definition of a distributed system** and illustrate its main **characteristic**?
2. What is "**Middleware**" and why is it often called the "middle helper"(write the main points)?
3. What is the main difference between **Centralized** systems - **Decentralized** systems - **Distributed** systems (support your answer with drawing)
4. Give the reasons of **why the Distributed Systems are needed?**
5. According to **resource sharing** in distributed system, write the **types of resources** and the **benefit of sharing**?
6. What does it mean for a distributed system to be "**scalable**" and "**fault tolerance** "?
7. What are the main differences between a **single** computer application and a **distributed application**?
8. Illustrate why we consider the **Facebook application** is an example of distributed systems (support your answer with its main components and how the components work together) ?

I hope you answer all the questions with my best wishes